



Two-stage graph estimation for complex motifs

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ABSTRACT

In this article, I develop an Induced-Sequential-Sampling (ISS) algorithm for subgraph estimation and propose a two-stage graph estimation strategy for verifying complex motifs in large and directed networks. I also derive the asymptotic properties of the estimators for subgraph identification and motif verification, and the simulation results are consistent with expectations. Finally, I provide an example for demonstrating how the proposed framework can be applied to investigate fragmentation in real-world trade networks.

1. Introduction

Modern graph statistics has made substantial progress. Graph estimation makes it possible to uncover macro- or micro-level information latent in networks (Hardiman and Katzir, 2013; Cohen et al., 2014); some existing studies have also combined statistical inference and reached toward causal inference, including identifying graph generation based on Exponential Random Graph Models (ERGM) and conducting subgraph inference in the presence of errors within networks (Cranmer and Desmarais, 2011; Chang et al., 2022). From another “direction”, recent work is also able to start from micro-level information and recover real networks in reverse (Xing et al., 2025). Nevertheless, one concept deserves closer scrutiny—as an important concept in complex network analysis, “motif” is a good tool for us to understand the meso-scale structures of networks (Xiao et al., 2025). For instance, motifs help identify signaling pathways in gene regulation in biological networks, analyze information diffusion in social networks, and, in economics and politics, measure an increasingly important new concept, namely, “fragmentation”.

A “motif” can be defined as a subgraph with a specific connection pattern (Wang et al., 2014), or, from a statistical perspective, as a subgraph that appears significantly more frequently than expected in random networks (Picard et al., 2008). Fig. 1 shows several connection patterns of subgraphs. The existing literature has made some progress in subgraph identification. Some current algorithms can efficiently identify simple subgraphs (Wang et al., 2018; Bressan et al., 2018; Ribeiro et al., 2022), while other studies identify motifs by analyzing the sampling distribution of subgraphs (Picard et al., 2008; Micale et al., 2018). However, the motivation of this article stems from two limitations. First, mainstream algorithms remain insufficiently effective in handling complex subgraphs (more than 5 vertices), and are difficult to apply to directed networks. Second, existing statistical methods usually require the exact observed count of subgraphs.

Therefore, this article inherits the idea of combining graph estimation with statistical inference, namely, “graph inference”, and applies it to the analysis of motifs in complex networks. I make two main contributions: (1) I develop an Induced-Sequential-Sampling (ISS) algorithm to estimate the count of any specified complex subgraph in any network (simple/weighted undirected/directed cases); (2) I construct a two-stage identification strategy, in which subgraph estimators are constructed separately for the observed network and the null model with corresponding statistical tests being developed. In addition, I provide a complete set of executable .R code for implementing the above estimation procedure. Finally, I present the asymptotic properties of the two-stage estimators and use a simple case to illustrate how they can be applied to identifying “fragmentation” in international trade.

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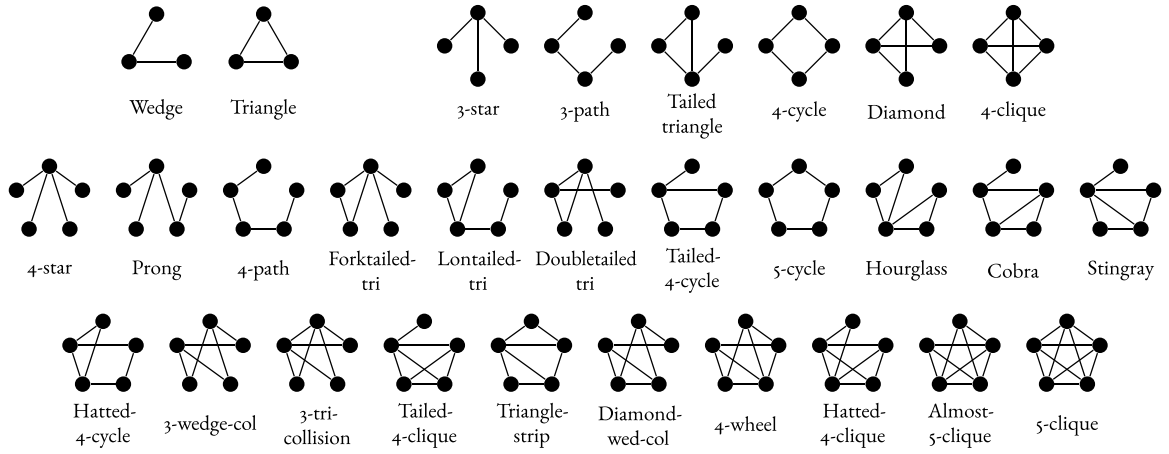


Fig. 1. All 29 connected subgraph patterns of 3-, 4-, and 5-vertex.

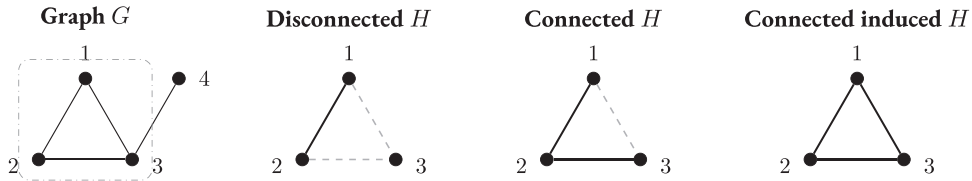


Fig. 2. Subgraph and connected induced subgraph.

2. Two-stage graph estimation strategy

2.1. Notation and definition

We begin with the simplest case to clarify and delimit several basic concepts and notations. Here, the 1-mode undirected network is treated as the default case. Further details and corresponding directed cases will be discussed in the Supplementary Information. Formally, denote 1-mode undirected network by the simple undirected graph $G = (V, E)$, and the connected induced subgraph consists of vertices of G as $H = (V_H, E_H \mid V_H \subset V, E_H \subset E)$ (presented in Fig. 2); then, we can provide the statistical definition for motif:

Definition 1 (Motif). Under a random-network population distribution $\mathcal{G}_0$, and a significance level α , if the right-tail probability satisfies $\mathbb{P}\{N_H(G^*) \geq N_H(G) \mid G^* \sim \mathcal{G}_0\} \leq \alpha$, then H is called a (genuine) motif in G .

We now define the mapping of an induced subgraph into the original graph G :

Definition 2 (Labeled Induced Embedding). Fix a labeling $S = (s_1, s_2, \dots, s_k)$ of the vertex set of the target subgraph H ; let $\phi : V_H \hookrightarrow V_G$ be an injection of the vertex set V_H into V_G , satisfying $(\phi(s_1), \dots, \phi(s_k)) = (v_1, \dots, v_k)$, where $v_1, \dots, v_k$ denote distinct vertices in G . If for $\forall a, b \in S$, the induced constraint $A_H(a, b) = A_G(\phi(a), \phi(b))$ holds, then ϕ is called a “labeled induced embedding” of the target subgraph H in the graph G .

On the basis of Definition 2, let the set of all labeled induced embeddings be denoted by $\text{Emb}_{\text{ind}}(H, G) = \{\phi : V_H \hookrightarrow V_G : G[\phi(V_H)] \cong_{\text{ind}} H\}$, i.e., the set of all injections under which the subgraph induced by $\phi(V_H)$ is isomorphic to H , and define the indicator function $I_H(G; \phi) = \mathbb{1}\{A_G(\phi(a), \phi(b)) = A_H(a, b) \mid \forall (a, b) \in V_H^2\}$; then, the total number of labeled induced embeddings of the target subgraph H can be expressed as:

$$L_H(G) = \sum_{\phi \in \text{Inj}([k], V_G)} I_H(G; \phi) = |\text{Emb}_{\text{ind}}(H, G)| \quad (1)$$

where $\text{Inj}([k], V_G)$ denotes the set of all injections from the labeled vertices S to V_G . Then, we have the following corollary:

Corollary 1 (Unlabeled Count). The set of all unlabeled induced embeddings is the quotient set of the labeled embeddings by the automorphism group $\text{Aut}(H)$, and the count of the target subgraph will be $N_H(G) = |\text{Aut}(H)|^{-1} L_H(G)$.

2.2. Stage 1: Subgraph estimation in targeted graph

The objective of the first-stage estimation is to estimate the count of the target subgraph in a given network G , i.e., $N_H(G)$. Enumeration will be almost infeasible for large networks, because its time complexity is $O(n^k)$. A natural idea is, therefore, “estimation”. Thus, I develop the induced-sequential-sampling (ISS) algorithm, following the idea of path sampling (Jha et al., 2015; Wang et al., 2018).

First, given the target subgraph H and the density of the target network $\hat{p}_G = \binom{n}{2}^{-1} |E|$, we can define the edge-constraint as $w_E = -\log(\hat{p}_G)$ and non-edge-constraint weights as $w_0 = -\log(1 - \hat{p}_G)$. Second, for a subgraph vertex u , we can define its global and conditional constraint score:

$$\begin{cases} \mathfrak{N}(u) = \sum_{v \neq u} [H_{uv} w_E + (1 - H_{uv}) w_0] \\ \mathfrak{C}(u | S) = \sum_{s \in S} [H_{us} w_E + (1 - H_{us}) w_0] \end{cases} \quad (2)$$

The first sampling step selects the vertex with the highest global constraint score $\mathfrak{N}(u)$, and each subsequent step selects, from the unsampled set S , the vertex with the highest conditional constraint score $\mathfrak{C}(u | S)$. Third, we begin to construct candidate vertex sets. Let the constraint ordering of the target subgraph be $O = (o_1, o_2, \dots, o_k)$. At step t , suppose that the selected target-network vertices are $X = (x_1, x_2, \dots, x_{t-1})$, where x_s corresponds to subgraph vertex o_s . The candidate set consists of all vertices that have not yet been sampled and that satisfy the induced constraints relative to the already sampled vertices:

$$C_t(\phi) = \{v \in V_\# : G[v, x_j] = H[o_t, o_j], \forall j < t\}. \quad (3)$$

where $V_\# = V \setminus \{x_1, \dots, x_{t-1}\}$ denotes the vertex set after removing vertices $(x_1, \dots, x_{t-1})$. In implementation, denote $N(x_s)$ as the neighbor vertex set of x_s and $\bar{N}(x_s)$ as the non-neighbor vertex set of x_s ; $N(x_s)$ and $\bar{N}(x_s)$ can be stored as logical bitsets, and the candidate set $C_t(\phi)$ can be obtained by successive intersections:

$$C_t = V_\# \cap \left(\bigcap_{s < t: H_{o_t, o_s}=1} N(x_s) \right) \cap \left(\bigcap_{s < t: H_{o_t, o_s}=0} \bar{N}(x_s) \right) \quad (4)$$

At any step of a sequential sampling process, if $C_t = \emptyset$, the path weight of that entire sampling will be 0; otherwise x_t is sampled uniformly from C_t . The weight W of a successful sampling path is the product of $|C_t|$ over all steps, i.e., $W = \prod_{t=1}^k |C_t|$. Then, we have the following corollary:

Corollary 2 (Path Weight). W is exactly the inverse probability of the sampled path.

Let $\{W_1, W_2, \dots, W_B\}$ be the path weights obtained from B independent sampling processes, we define the following Horvitz-Thompson estimator:
$$\begin{cases} \hat{L}_H(G) = \frac{1}{B} \sum_{b=1}^B W_b \\ \hat{N}_H(G) = \frac{\hat{L}_H(G)}{a_H} \end{cases},$$
 and given the target network and the target subgraph, we have the following proposition:

Proposition 1 (Unbiasedness). Both the labeled and unlabeled ISS estimators are unbiased, i.e., $\mathbb{E}[\hat{L}_H] = L_H$ and $\mathbb{E}[\hat{N}_H] = N_H$.

Now, consider the sampling asymptotic properties. Under G and H with given scales, while the sampling draws $B \rightarrow \infty$, we show that:

Proposition 2 (Consistency). Since W_b are i.i.d. and bounded in a finite network, $\hat{N}_H(G)$ is a consistent estimator of $N_H(G)$. The sample variance $s_{W'}^2 = (B-1)^{-1} \sum_b (W_b - \hat{L}_H)^2$ is also a consistent estimator of σ_H^2 .

Proposition 3 (Asymptotic Normality). If $\sigma_H^2 = \mathbb{V}(W_b) < \infty$, the Central Limit Theorem (CLT) gives: $\sqrt{B} [\hat{N}_H(G) - N_H(G)] \overset{\text{approx}}{\sim} N(0, \sigma_H^2 / a_H^2)$. Consequently, $(a_H \sqrt{B})^{-1} s_{W'}$ is the Monte Carlo standard error of the unlabeled count estimator.

2.3. Stage 2: Subgraph estimation in null model

The objective of the second-stage estimation is to estimate the expected size of the target subgraph under a random-network distribution, i.e., $\mathbb{E}[N_H(G^*)]$. A general approach is to generate a sequence of random networks by B times bootstrap, estimate the target subgraph count $\hat{N}_H(G^*)$ in each random network G^* , and then $B^{-1} \sum_b \hat{N}_H(G_b^*)$ will be an unbiased estimator of T_{G_0} . However, the randomness of such an estimator comes not only from the generation of random networks G^* , but also from the ISS sampling error conditional on each given G_b^* . We can therefore consider a conditional Monte Carlo (CMC) strategy (or so-called a generalized Rao-Blackwellization strategy without requiring the conditioning variable to be sufficient), i.e., in the second stage, we can directly estimate the same target-subgraph expectation $\mu_0 = \mathbb{E}[N_H(G^*)]$ under an independent-edge null model.

The null model $\mathcal{G}_0$ for a given graph is given by a probability matrix $P \in [0, 1]^{n \times n}$, which likewise satisfies: (1) symmetry, i.e., $p_{ij} = p_{ji}$; and (2) zero diagonal entries, i.e., $p_{ii} = 0$. Edges e_{ij} in the null model are generated independently. Two types of generations for random networks are usually available: (1) Erdős–Rényi type, for which $p_{ij} = \hat{p}_G$ for all $i < j$ in the given graph; and (2) Chung–Lu type, for which $p_{ij} = \min \left\{ \frac{d_i d_j}{2m}, 1 - \varepsilon \right\}$, where d_i and m denote the original undirected degree and the number of edges (Fasino et al., 2019).

For a given embedding ϕ , whether that embedding appears in a particular random network G^* is represented by a Bernoulli indicator variable $I_\phi(G^*)$. Under the population distribution $\mathcal{G}_0$ of the null model, the conditional Monte Carlo strategy allows us to consider the occurrence probability $p_H(\phi; \mathcal{G}_0) = \mathbb{E}[I_\phi(G^*)]$. The expected number of labeled occurrences of the target induced subgraph H , denoted by $M_H(\mathcal{G}_0)$, can then be computed as:

$$M_H(\mathcal{G}_0) = \mathbb{E}_{\mathcal{G}_0} [L_H(G^*)] = \mathbb{E}_{\mathcal{G}_0} \left[\sum_{\phi} I_\phi(G^*) \right] = \sum_{\phi} \mathbb{P}_{\mathcal{G}_0} [\phi \text{ forms } H] \quad (5)$$

where $\mathbb{P}_{\mathcal{G}_0}[\phi \text{ forms } H]$ is the probability that a given induced embedding ϕ forms the target induced subgraph under the null model, which can be expressed as:

$$\mathbb{P}_{\mathcal{G}_0}[\phi \text{ forms } H] = \prod_{\{a,b\}: a < b} p_{\phi(a), \phi(b)}^{H_{a,b}} (1 - p_{\phi(a), \phi(b)})^{1-H_{a,b}} \quad (6)$$

The unlabeled expectation is therefore $\mu_H(\mathcal{G}_0) = a_H^{-1} M_H(\mathcal{G}_0)$. Here, we again use the ISS estimator at this stage. Let the ordering of subgraph vertices be $o = (o_1, o_2, \dots, o_k)$. At step t , the mapping of $(o_1, o_2, \dots, o_{t-1})$ has been completed, and the probability contribution associated with drawing candidate vertex x is denoted by $g_t(x)$:

$$g_t(x) = \prod_{s < t} p_{x, x_s}^{H_{o_t, o_s}} (1 - p_{x, x_s})^{1-H_{o_t, o_s}} \quad (7)$$

Let $Z_t = \sum_{x \notin (x_1, \dots, x_{t-1})} g_t(x)$, and draw $x_t = x$ with probability $q_t(x) = \frac{g_t(x)}{Z_t}$, the log weight of a successful sampling path will be

$$\log W_0 = \sum_t \log Z_t. \text{ Then, we can still define the following Horvitz-Thompson estimator: } \begin{cases} \hat{M}_H(\mathcal{G}_0) = \frac{1}{R} \sum_{r=1}^R W_{0,r} \\ \hat{N}_H(\mathcal{G}_0) = \frac{\hat{M}_H(\mathcal{G}_0)}{a_H} \end{cases}, \text{ where } W_{0,r} \text{ denotes}$$

the path weight obtained from the r th independent sampling, and the properties of these two estimators are the same as those stated in Proposition 1–3.

2.4. Motif verification

So far, we have obtained both estimates in the given and random networks. We now consider statistically testing via statistical construction, to examine whether target subgraphs in the targeted network occur significantly more frequently than in the null model.

Denote $\theta = N_H(G)$ as the subgraph count in the real network, $\mu_0 = \mathbb{E}[N_H(G^*)]$ as the null-model expectation, and $\tau_0^2 = \mathbb{V}[N_H(G^*)]$ as the null-model variance. Here we construct the log ratio $\beta = \log \left(\frac{\theta}{\mu_0} \right)$, and refer to it as the subgraph enrichment coefficient β_{enr} . The corresponding estimator is $\hat{\beta} = \log \left(\frac{\hat{\theta}_B}{\hat{\mu}_{0,R}} \right)$, which can be computed from $\hat{\theta}_B$ and $\hat{\mu}_{0,R}$, obtained respectively from the two stages of estimation. Then, we have the following propositions:

Proposition 4 (Consistency). $\hat{\beta}$ is a consistent estimator of β , and $s_{\hat{\beta}}^2 = \frac{s_{\theta}^2}{B\hat{\theta}_B^2} + \frac{s_{\mu}^2}{R\hat{\mu}_{0,R}^2}$ is also a consistent estimator of $\mathbb{V}(\hat{\beta})$.

Proposition 5 (Asymptotic Normality). $\hat{\beta} - \beta$ is asymptotically normally distributed.

For a population null-model distribution $\mathcal{P}_0$, any null random network G^* follows the null-model distribution, i.e., $G^* \sim \mathcal{P}_0$. The true number of targeted subgraphs in the random network is $N_H(G^*)$, following the distribution $\mathcal{L}\{N_H(G^*) : G^* \sim \mathcal{P}_0\}$. The true number of targeted subgraphs in the random network has expectation $\mu_0 = \mathbb{E}[N_H(G^*)]$ and variance $\tau_0^2 = \mathbb{V}[N_H(G^*)]$. Meanwhile, we assume that the subgraph count under the null model can be approximated by a Poisson distribution:

Assumption 1 (Poisson Approximation). $N_H(G^*) \stackrel{\text{approx}}{\sim} \text{Poisson}(\mu_0)$, so that $\tau_0^2/\mu_0 \rightarrow 1$.

To construct a simple z-test, normality is required, which, however, requires $\mu_0 \rightarrow \infty$. Here, we therefore impose the following finite-sample approximation:

Assumption 2 (Poisson-to-Normal Approximation). In finite samples, μ_0 is sufficiently large to be approximated by a normal distribution, i.e., $N_H(G^*) \stackrel{\text{approx}}{\sim} N(\mu_0, \mu_0)$ (In practice, the normal approximation can be used since $\mu_0 \geq 15$, see: Johnson et al. (2005)).

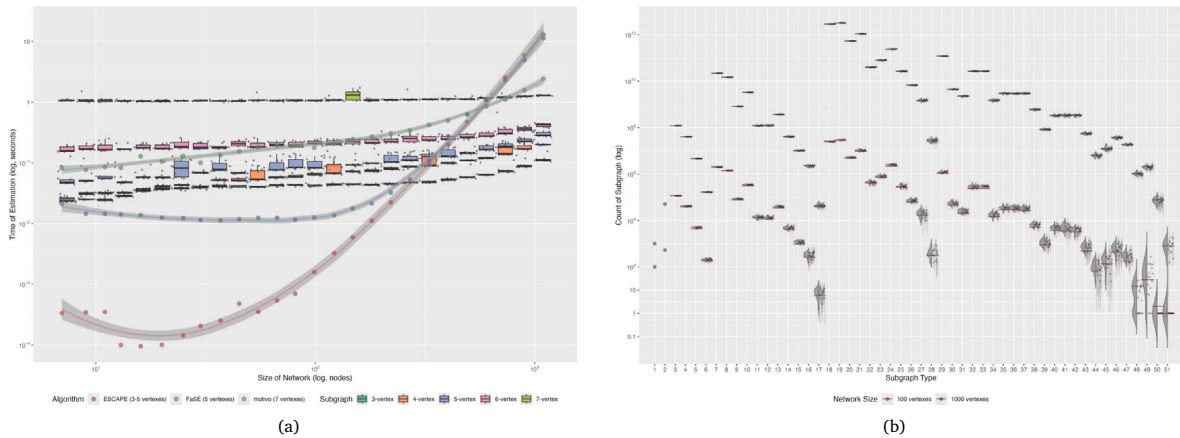


Fig. 3. Computational cost and accuracy of graph estimation.

Now, we can construct a z-statistic and test whether $\hat{\beta}$ is significantly greater than zero, thereby determining whether a particular subgraph can be regarded as a genuine motif. Specifically, we have the following proposition:

Proposition 6 (z-Statistic). Under Assumptions 1–2, for the null hypothesis $H_0 : G \sim \mathcal{P}_0$, equivalently $N_H(G) = \mathbb{E}[N_H(G^*)]$, and the right-tail alternative $H_1 : N_H(G) > \mathbb{E}[N_H(G^*)]$, construct the z-statistic $Z_\beta = \left[s_\theta^2(B\hat{\theta}^2)^{-1} + s_\mu^2(R\hat{\mu}_0^2)^{-1} + \hat{\mu}_0^{-1} \right]^{-1/2} \log \left(\hat{\theta}/\hat{\mu}_0 \right)$. Then $Z_\beta \stackrel{\text{approx}}{\sim} N(0, 1)$.

Nonetheless, the z-test above relies on the Poisson approximation and on the tail behavior under the induced constraints—e.g., for cliques, dense, or highly overlapping subgraphs, their count within a graph may generally deviate from the Poisson distribution. Therefore, we further consider random network calibration based on bootstrap sampling. The procedure is as follows: First, generate K i.i.d. random networks from the null-model $\mathcal{G}_0$, i.e., $G_1^*, \dots, G_K^* \stackrel{i.i.d.}{\sim} \mathcal{G}_0$. Second, for each random network, estimate the target subgraph count using the ISS algorithm, i.e., $\hat{\theta}_k^* = \hat{N}_H(G_k^*)$. Then, using the same direct null-expectation estimator $\hat{\mu}_0$, construct the subgraph enrichment coefficient estimator for each random network, i.e., $\hat{\beta}_k^* = \log \left(\frac{\hat{\theta}_k^*}{\hat{\mu}_0} \right)$, and take the mean $\bar{\beta}^* = K^{-1} \sum_{k=1}^K \hat{\beta}_k^*$. Meanwhile, define the sample variance $\hat{\omega}_\beta^2 = K^{-1} \sum_{k=1}^K \left(\hat{\beta}_k^* - \bar{\beta}^* \right)^2$. We have the following propositions:

Proposition 7 (Consistency). $\bar{\beta}^*$ is a consistent estimator of $\mathbb{E}[\beta^*] = \mathbb{E} \left[\log \left(\frac{N_H(G^*)}{\mu_0} \right) \right]$, and $\hat{\omega}_\beta^2$ is also a consistent estimator of $\mathbb{V}[\beta^*] = \mathbb{V} \left[\log \left(\frac{N_H(G^*)}{\mu_0} \right) \right]$.

Proposition 8 (Asymptotic Normality). Construct the z-statistic $Z_\beta^* = \left[s_\theta^2(B\hat{\theta}^2)^{-1} + s_\mu^2(R\hat{\mu}_0^2)^{-1} + \hat{\omega}_\beta^2 \right]^{-1/2} \log \left(\frac{\hat{\theta}}{\hat{\mu}_0} \right)$ by employing $\hat{\omega}_\beta^2$ as the plug-in estimator. Then, $Z_\beta^* \stackrel{\text{approx}}{\sim} N(0, 1)$.

3. Data simulation and application in international trade

I first evaluate the subgraph estimation algorithm. I compare the computational cost of the proposed algorithm with mainstream ones, including Paredes and Ribeiro (2015), Pinar et al. (2016), and Bressan et al. (2018), in networks with fewer than 1000 vertices, and present the results in Fig. 3(a). The proposed algorithm performs better for medium- and large-scale networks, capable of estimating more complex subgraphs (including directed cases). I then compare the accuracy with exact approaches (based on 1000 samples), with the results shown in Fig. 3(b). The estimated results are almost consistent with exact counting and yield narrow confidence intervals, since the target subgraph occurs frequently in the network.

I next evaluate the two-stage estimation procedure. I construct non-random networks according to specific rules and compare the motif verification results for several highly symmetric motifs. In brief, for any graph density $\hat{p}_G$ and a given k -vertex target subgraph (with e_k edges), the constructed n -vertex network satisfies at least $\hat{p}_G^{e_k} \binom{n}{k} \geq 20$ (i.e., the count expectation of targeted subgraphs is no less than 20). The results are reported in Table 1, which shows that at lower graph density, the Poisson approximation yields relatively high consistency with the empirical p -value obtained from bootstrap sampling as the benchmark; while at higher graph density, the Poisson approximation gives more optimistic p -values, in which case random-network calibration performs better.

Finally, I apply the two-stage procedure to international trade networks as an illustrative case. I first construct simple undirected networks by calculating the Revealed Comparative Advantage (RCA) indices. Here, I distinguish three cases: (1) impose no

Table 1
Subgraph and Motif verification results.

Dens.	Erdős-Rényi null model				Chung-Lu null model			
	β	p (Pois.)	p (cal.) ^a	p (emp.) ^b	β	p (Pois.)	p (cal.)	p (emp.)
 (triangle) ~ $\hat{p}_G = 0.001$ $\hat{p}_G = 0.1$ $\hat{p}_G = 0.5$	0.528***	[0.008]	[0.009]	[0.009]	0.244	[0.098]	[0.110]	[0.105]
	0.620**	[0.004]	[0.035]	[0.036]	0.333	[0.047]	[0.163]	[0.161]
	0.335*	[<0.001]	[0.070]	[0.069]	0.380**	[<0.001]	[0.044]	[0.046]
 (4-clique) ~ $\hat{p}_G = 0.01$ $\hat{p}_G = 0.1$ $\hat{p}_G = 0.5$	0.344**	[0.044]	[0.049]	[0.046]	0.357**	[0.038]	[0.040]	[0.040]
	0.515*	[0.009]	[0.053]	[0.055]	0.472*	[0.013]	[0.073]	[0.072]
	0.859**	[<0.001]	[0.042]	[0.043]	0.739*	[<0.001]	[0.066]	[0.065]
 (5-clique) ~ $\hat{p}_G = 0.05$ $\hat{p}_G = 0.1$ $\hat{p}_G = 0.5$	0.498***	[0.007]	[0.010]	[0.009]	0.489**	[0.007]	[0.010]	[0.010]
	0.391*	[<0.001]	[0.059]	[0.053]	0.398**	[0.019]	[0.050]	[0.049]
	1.979**	[<0.001]	[0.026]	[0.028]	1.693**	[<0.001]	[0.044]	[0.045]
 (6-clique) ~ $\hat{p}_G = 0.05$ $\hat{p}_G = 0.1$ $\hat{p}_G = 0.5$	0.761**	[0.038]	[0.038]	—	0.760**	[0.038]	[0.038]	—
	0.338*	[0.061]	[0.074]	[0.074]	0.214	[0.158]	[0.174]	[0.175]
	1.768*	[<0.001]	[0.066]	[0.068]	1.186	[<0.001]	[0.146]	[0.147]

*** denote $p < 0.01$.

** denote $p < 0.05$.

* denote $p < 0.1$.

^a Here, the number of random networks used for calibration is 50.

^b The bootstrap time for calculating the empirical p -values is 1000.

RCA threshold, for examining the global structure; (2) investigate the “dominant structure” represented by $RCA > 1$, for which I set $\begin{cases} e_{ij} = 1, & RCA > 1 \\ e_{ij} = 0, & RCA \leq 1 \end{cases}$; and (3) investigate the “residual structure” represented by $0 < RCA \leq 1$, for which I set $\begin{cases} e_{ij} = 1, & 0 < RCA \leq 1 \\ e_{ij} = 0, & RCA \leq 0 \text{ or } RCA > 1 \end{cases}$. Also, I have provided corresponding robustness checks in the SI. Second, I estimate the enrichment coefficients for triangle, 4-, and 5-cliques as target motifs from 2000 to 2023 (the null model used in motif verification here is the Erdős-Rényi-type null model). The results for the global structure show that the β etas of all three target motifs have indeed declined markedly over the past two decades. Meanwhile, such a decline is suggested to be mainly driven by the residual structure, whereas β for the dominant structure remains almost unchanged or even increases slightly. In a sense, this result can be interpreted as evidence of “fragmentation”—integration implies the formation of symmetric, stable, or dense trade patterns, so the number of k -cliques should be significantly larger than expected. Fragmentation, by contrast, implies that the occurrence of these trade patterns is no different from randomization. The result suggests the “residual structure”, which typically involves less-developed countries or emerging markets, mainly drives the increase in fragmentation (see Fig. 4).

4. Concluding remark

In this article, I develop a subgraph estimation algorithm and propose a two-stage graph estimation strategy. The simulation results suggest that both the algorithm and the estimation procedure have favorable asymptotic properties; and more critically, the two-stage estimation gives clear economic content and interpretation. This article is, of course, only a first step, but I believe that the proposed estimation framework has substantial scope for further development. First, extending the framework to multi-partite graphs, and potentially to hyper-graphs, would make it applicable to a broader set of economic issues. Second, inference on motifs may also help deepen our understanding of broader graph inference or graph-generating models (Chang et al., 2022; Xing et al., 2025). Third, graph inference may also prove useful for modern causal identification in economics, especially in estimators explicitly involving network terms (Cranmer and Desmarais, 2011; Bartolucci et al., 2024; Mao and Yang, 2026). Finally, further statistical work could refine the graph estimation developed in this article.

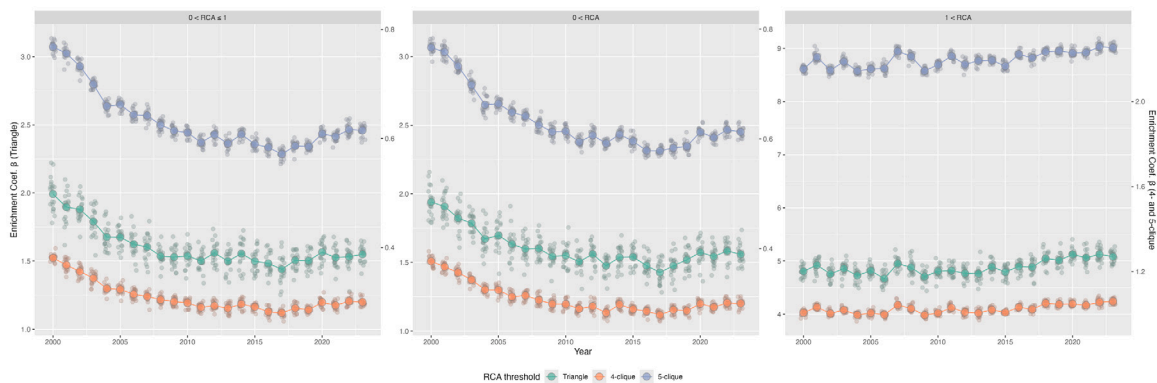


Fig. 4. Motif verification of international trade from 2000 to 2023.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.spl.2026.110958>.

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